Enterprise RAG Platform

An engineering research report on Hybrid retrieval (BM25 + dense) + reranking + eval harness

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2024

Contents

1. Abstract 3

2. Introduction 3

2.1 Research Problem . . . 3

2.2 Research Questions and Hypotheses . . . 3

3. Technical Background Review 4

3.1 Design Foundations . . . 4

3.2 Supporting Examples . . . 4

4. Research Method 4

5. Data Description 4

6. Analysis 5

6.1 Benchmark and Validation Results . . . 5

6.2 Test Result Screenshots . . . 5

6.3 Validation Checklist . . . 6

7. Discussion 6

8. Conclusion 7

9. Future Work 7

10. References 7

1. Abstract

This brief documents a retrieval-augmented generation core built for enterprise knowledge use cases where accuracy, traceability, and tenant boundaries matter more than surface-level demo output. The final stack turned retrieval into a measurable engineering system. Hybrid retrieval with reranking improved top-rank quality while preserving grounded, citation-backed answers and deterministic offline reproducibility. The implementation evidence is drawn from committed repository artefacts, benchmark outputs, automated tests, and generated screenshots.

Keywords: Hybrid retrieval (BM25 + dense) + reranking + eval harness, implementation evidence, benchmark results, project validation

2. Introduction

The operating problem was straightforward: teams keep policies, tickets, and wiki content in separate silos, but a useful assistant still has to retrieve the correct passage, cite it, and avoid leaking data across tenants. Production-grade retrieval-augmented generation - offline, deterministic, and evaluated.

2.1 Research Problem

The operating problem was straightforward: teams keep policies, tickets, and wiki content in separate silos, but a useful assistant still has to retrieve the correct passage, cite it, and avoid leaking data across tenants.

2.2 Research Questions and Hypotheses

Research question: Does the implemented project address the stated technical problem in a complete and reviewable manner?

Hypothesis: The repository design, architecture notes, and implementation artefacts are expected to demonstrate end-to-end coverage of the core project objective.

Research question: Do the benchmark and test artefacts support the main performance or quality claim?

Hypothesis: The recorded evidence is expected to support the headline result reported for this project: Hybrid+Rerank: Recall@1 0.92, nDCG@10 0.96.

Research question: Can the results be reviewed and reproduced from committed repository evidence?

Hypothesis: The presence of 28 automated tests, benchmark outputs, and generated screenshots is expected to provide a transparent validation path.

3. Technical Background Review

3.1 Design Foundations

1. Hybrid fusion was kept score-scale agnostic so lexical and dense paths can coexist honestly. Built around retrieval quality, citation traceability, and strict tenant isolation for enterprise knowledge systems.

2. Citation grounding was treated as a product requirement, not a dashboard extra.

3. Isolation is applied inside the retriever to prevent ranking leakage and wasted top-k budget.

3.2 Supporting Examples

- Streaming corpus generation and chunking with metadata.
- BM25, dense, hybrid retrieval, and learned reranking.
- Extractive synthesis with citation offsets.

4. Research Method

The project was reviewed as an engineering artefact using repository documentation, architecture notes, benchmark evidence, automated tests, and generated screenshots. Mapped enterprise search failure modes to measurable retrieval goals: recall, rank quality, groundedness, and tenant safety. Implemented chunking, BM25, dense retrieval, RRF fusion, and a logistic reranker behind clean interfaces. Benchmarked four retrieval paths on held-out questions and compared quality against latency and groundedness. Added deterministic seeds, artifact generation, and multi-tenant tests so the same run can be reviewed repeatedly.

5. Data Description

Source: committed repository artefacts including implementation code, automated tests, benchmark outputs, architecture notes, and generated visual evidence. Coverage: 28 automated tests, 1 benchmark table(s), and 4 screenshot asset(s) were available for document preparation. Schema / artefact types: README overview, ARCHITECTURE design note, benchmark markdown and CSV outputs, test suites, and figure assets generated from the project run path. Availability: all evidence cited in this document is sourced from the repository itself.

6. Analysis

6.1 Benchmark and Validation Results

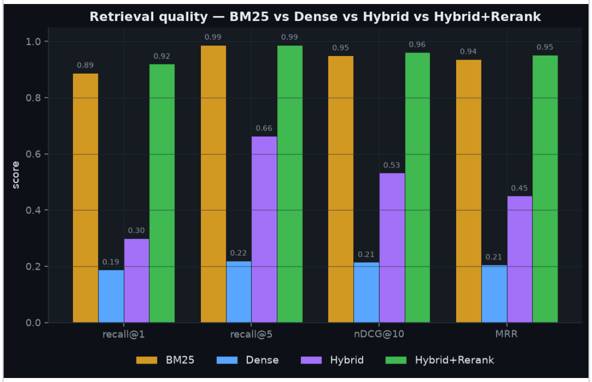
The headline repository result is reported as: Hybrid+Rerank: Recall@1 0.92, nDCG@10 0.96. The table below reproduces benchmark evidence included in the repository where available.

method	recall@1	recall@3	recall@5	recall@10	mrr
BM25	0.888	0.988	0.988	0.988	0.937
Dense	0.188	0.216	0.220	0.244	0.206
Hybrid	0.300	0.540	0.664	0.792	0.450
Hybrid+Rerank	0.920	0.984	0.988	0.988	0.952

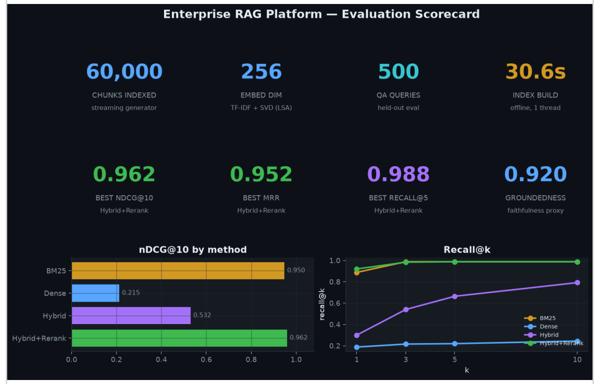
The scorecards show Hybrid+Rerank leading both BM25 and dense-only retrieval on top-rank quality. Latency remains operationally plausible because the reranker only touches a bounded candidate set. Groundedness stays stable across retrieval modes because synthesis remains extractive and citation-aware.

6.2 Test Result Screenshots

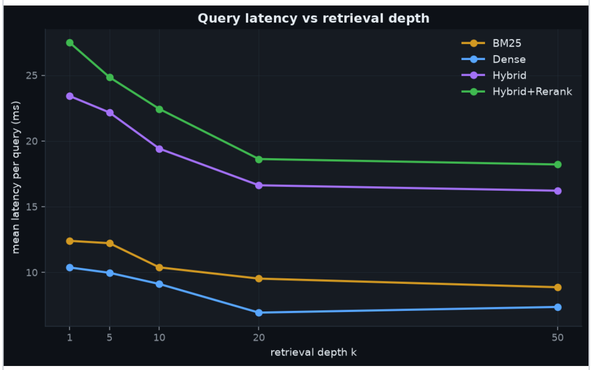
The following figures are drawn directly from the repository screenshot assets and are included as visual evidence of measured outputs, dashboards, or generated validation artefacts.



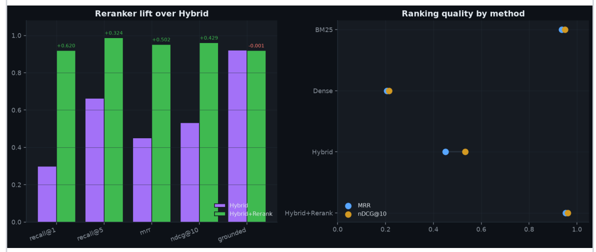
Retrieval quality - 4 methods compared



Evaluation scorecard



Query latency vs retrieval depth



Reranker lift & ranking quality

6.3 Validation Checklist

Item	Status	Evidence
Automated tests	Available	28 tests committed in repository validation flow.
Benchmark results	Available	Benchmark markdown tables and result summaries were reviewed.
Screenshots	Available	4 screenshot asset(s) included in the repository evidence set.
Architecture note	Available	ARCHITECTURE.md documents design choices, scale assumptions, and trade-offs.

7. Discussion

The final stack turned retrieval into a measurable engineering system. Hybrid retrieval with reranking improved top-rank quality while preserving grounded, citation-backed answers and deterministic offline reproducibility. The scorecards show Hybrid+Rerank leading both BM25 and dense-only retrieval on top-rank quality. Latency remains operationally plausible because the reranker only touches a bounded candidate set. Groundedness stays stable across retrieval modes because synthesis remains extractive and citation-aware. The analysis indicates that the repository is strongest where implementation, benchmark artefacts, and screenshots point to the same conclusion.

8. Conclusion

This project document concludes that Enterprise RAG Platform is best understood as a complete technical implementation supported by repository-native evidence. This brief documents a retrieval-augmented generation core built for enterprise knowledge use cases where accuracy, traceability, and tenant boundaries matter more than surface-level demo output.

9. Future Work

- Swap the deterministic embedder for a production encoder behind the same interface.
- Add online evaluation slices for business unit, tenant tier, and corpus freshness.
- Move from in-memory retrieval to shard-aware ANN infrastructure for larger corpora.

10. References

1. README.md - primary repository overview and headline results.
2. ARCHITECTURE.md - system design, implementation rationale, and scale notes.
3. benchmarks/latency.csv - benchmark or result artefact used in the analysis section.
4. benchmarks/results.json - benchmark or result artefact used in the analysis section.
5. benchmarks/retrieval_quality.csv - benchmark or result artefact used in the analysis section.
6. benchmarks/retrieval_quality.md - benchmark or result artefact used in the analysis section.
7. tests/ - automated validation suite used to support implementation claims.
8. assets/latency_vs_k.png - generated screenshot evidence included in the report.